

Red5-AHU

One chart to keep buildings comfortable, healthy, and energy-smart

A plain-language look at how a smart air-handler tool meets every major ASHRAE standard — and why every Building Management System needs it.

Edge-native

No cloud

BACnet-native

ASHRAE-aligned

THE CHALLENGE

Three things every building must do — at once

Feel comfortable

Not too hot, not too cold, not too humid.

Stay healthy

Enough fresh outside air so the air is safe to breathe.

Save energy

Do both of the above without wasting power or money.

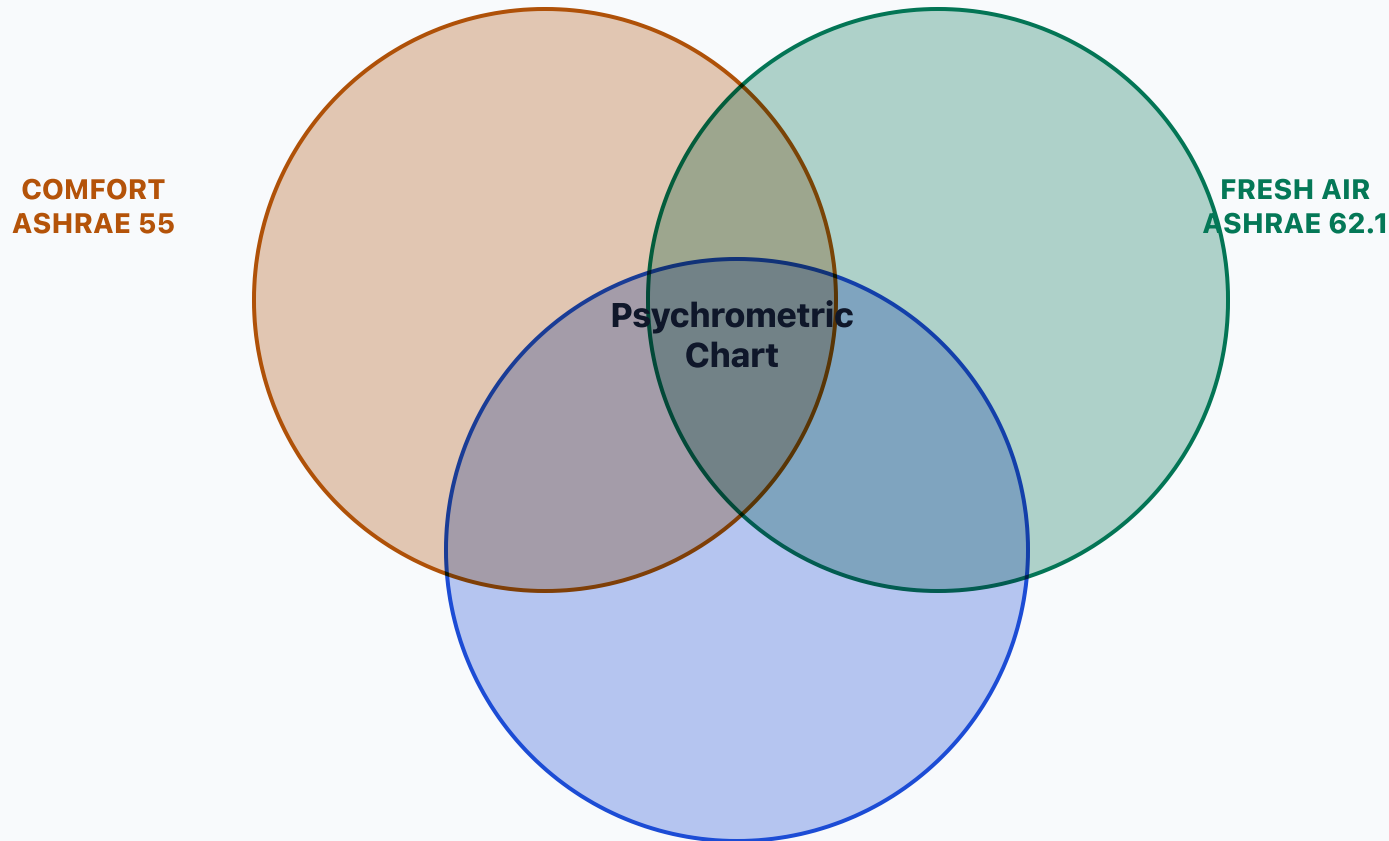
The catch

These goals fight each other. More comfort and more fresh air usually cost more energy. Getting all three right, all day, in every season — is hard.

Four ASHRAE standards, in one line each

ASHRAE 55	Comfort	Will people feel comfortable?
ASHRAE 62.1	Fresh air	Is there enough outside air to stay healthy?
ASHRAE 90.1	Energy	Are we doing it without wasting energy?
Guideline 36	The referee	How do we run the equipment to hit all three?

One picture where all four rulebooks meet



Why it works

All four standards live on the SAME chart — temperature across the bottom, humidity up the side.

Plot one dot for the air and you can instantly see comfort, fresh-air mix, and energy — all together.

Guideline 36 is the referee — the control logic that keeps the dot in the sweet spot automatically.

One dot on the chart answers every question

55 — Comfort

The comfort zone is drawn on the chart. Is the dot inside the good temperature/humidity band? If not, which way is it off?

62.1 — Fresh air

The mixed-air dot sits on the line between return and outside air — its position shows the fresh-air share at a glance.

90.1 — Energy

Enthalpy lines reveal free cooling: when outside air is cooler, the economizer meets the load with no mechanical cooling.

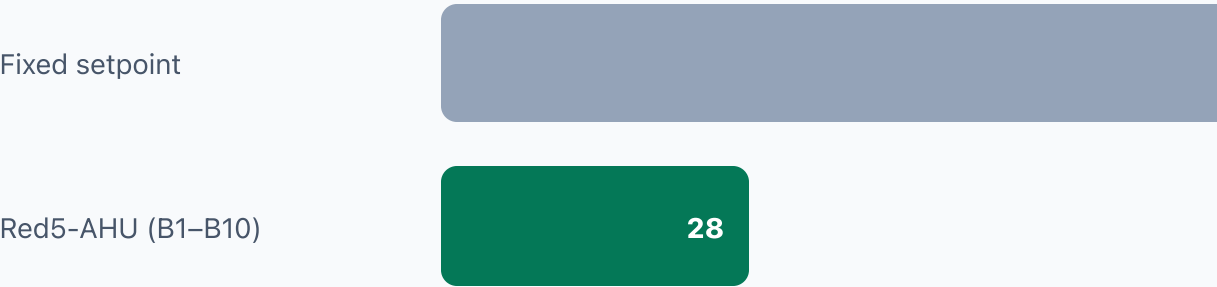
G36 — Sequences

Visual proof the automatic sequences land the dot in the comfort zone, on the fresh-air line, at the lowest-energy path.

In one sentence: plot where the air IS, see where it MUST be, how it gets there, and whether it does so at minimum energy.

Red5-AHU vs a traditional BMS

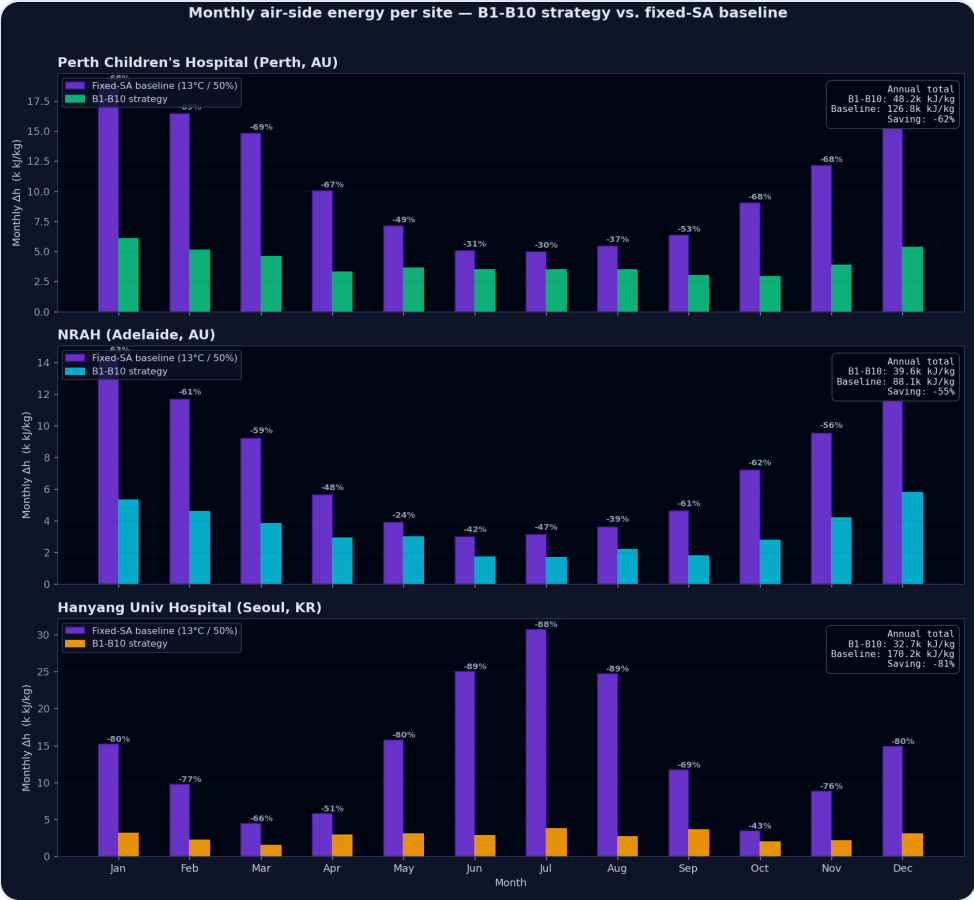
Supply-air energy (lower is better)



≈ 70–75% less supply-air energy than a fixed year-round setpoint (internal estimate, temperate climate).

	TRADITIONAL BMS	RED5-AHU
Where it runs	Cloud / head-end PC	On the controller
Standards built in	Add-on / manual	55·62.1·90.1·G36
Diagnose an AHU	Dig through trends	One glance
Vendor lock-in	Usually yes	Open protocols
Languages	Often one	Five

The control follows the seasons



What you're looking at

Instead of one fixed target all year, Red5-AHU sets the supply-air target to match the season (climate bands B1–B10).

In mild months it leans on free outside-air cooling; in harsh months it does only as much conditioning as needed.

The result: month after month the air handler uses far less energy than a fixed setpoint — automatically, with no operator effort.

Why it's an essential BMS tool

Sees everything at once

Comfort, fresh air, and energy on one live picture — no guesswork, no digging through data.

Saves real energy

Automatic free-cooling and climate-adaptive supply air cut supply-air energy ~70%.

Runs on the controller

No cloud, no extra server. Fewer parts to fail; keeps working if the network drops.

Speaks BMS natively

BACnet-native and open (MQTT/Modbus) — no vendor lock-in; integrates with what you have.

Meets the standards

Aligned to ASHRAE 55 / 62.1 / 90.1 and Guideline 36 out of the box.

Easy to roll out

Diagnose at a glance, clone the setup to the next unit, in five languages.

Four rulebooks. One chart. Zero cloud.

Red5-AHU turns every air handler into a self-diagnosing, self-optimizing asset that satisfies ASHRAE 55, 62.1, 90.1 and Guideline 36 — right on the controller.